



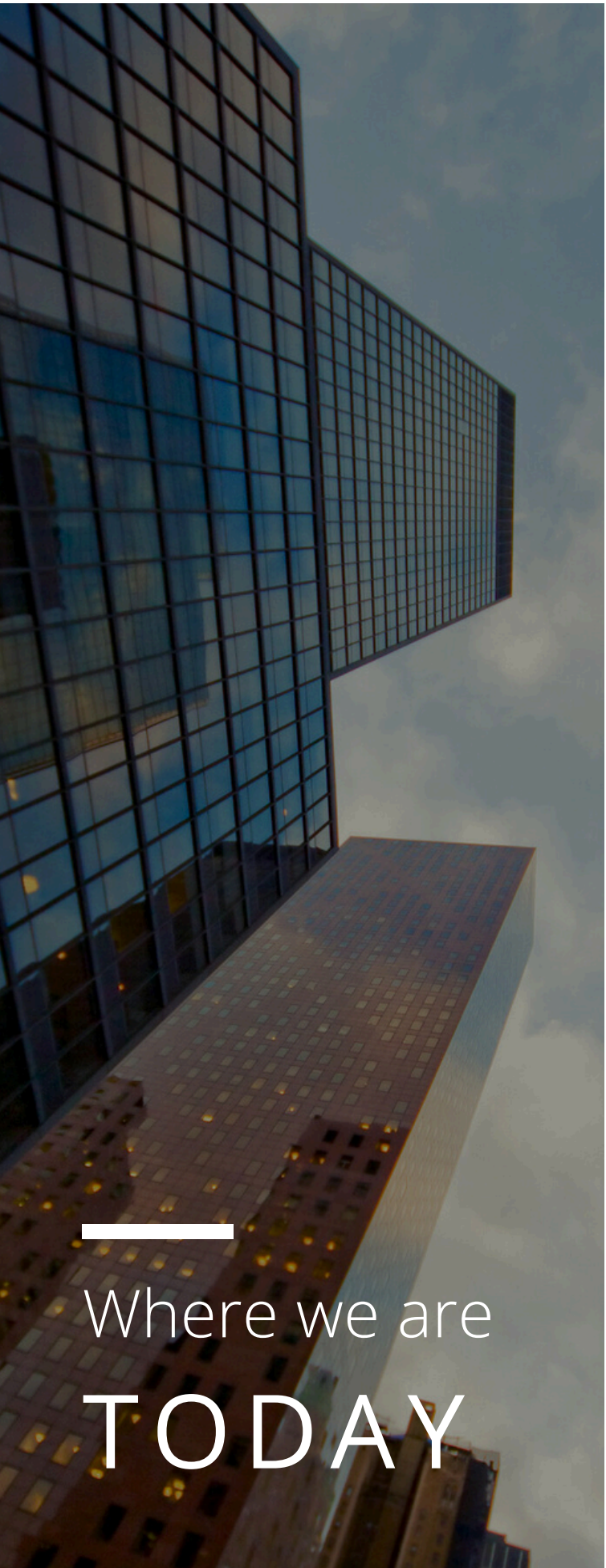
NETWORK NUTS

CERTIFIED

KUBERNETES ADMINISTRATOR

2026

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Certified Kubernetes Administrator

OVERVIEW

The Certified Kubernetes Administrator (CKA) certification is designed to ensure that certification holders have the skills, knowledge, and competency to perform the responsibilities of Kubernetes Administrators.

The CKA certification allows certified administrators to quickly establish their credibility and value in the job market, and also allowing companies to more quickly hire high-quality teams to support their growth.

Kubernetes has evolved from a simple container orchestrator to a comprehensive platform that powers modern cloud-native applications.

Kubernetes skills are the most in demand for modern operational roles, such as devops and site reliability engineering (SRE) jobs. These are also the roles where the money is.

Where we are
TODAY

Course Outline

- Introduction to Kubernetes
 - Introduction to Certified Kubernetes Administrator Training
 - Need of Kubernetes as a Container Orchestrator
 - Components in Kubernetes cluster
- Learning Core Concepts of Kubernetes Cluster
 - Etcd in Kubernetes - stacked and external
 - Kube-API Server
 - Kube-Controller Manager
 - Kube Scheduler
 - Kubelet
 - Container Engine
 - Kube-proxy
 - **LAB:** Installing a Kubernetes cluster using kubeadm - stacked etcd
 - **LAB:** Installing Production grade Kubernetes cluster - external etcd
 - Exploring Cluster
 - How to Access Kubernetes Cluster in Production
 - **LAB:** Accessing Kubernetes Cluster from Remote Machine

Troubleshooting, discussions & production issues:



- Kubernetes cluster not able to communicate with external etcd
- Migrate workload to a new Kubernetes cluster with external etcd
- Scale your existing external etcd cluster
- Why in high performance Kubernetes clusters Cilium is preferred over Calico
- What checks you will perform before installing CNI (calico) on existing cluster

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Course Outline

- Managing Business Application in Pods
 - Difference between containers and pods
 - Understanding pod overhead
 - **LAB:** Creating pods imperatively
 - **LAB:** Creating pods declaratively using YAML
 - Kubectl create vs Kubectl apply
 - **LAB:** Versioned resources with kubectl apply
 - Labels and Selectors in a Pod
 - **LAB:** Settings labels on pods / resources
 - **LAB:** Using selectors to filter workload
 - Controlling compute using quality of service
 - Understanding besteffort, burstable and guaranteed quality of service
 - **LAB:** Setting Requests and Limits on Pods
 - Capturing resource metrics
 - **LAB:** Installing Metric Server
 - **LAB:** Filtering resources based on compute usage
 - **LAB:** Supply environment variable to containers

Troubleshooting, discussions & production issues:



- Pod getting ImagePullBack or ErrImagePull error
- Pod having CrashLoopBack error after it starts
- Pod status is Pending or FailedScheduling
- Pod is getting OOMKilled regularly
- Clients are complaining that application response time has increased
- What is pause container in the pod
- How events can be helpful troubleshooting workload issues
- You don't want cpu throttling on containers if cpu is available on node
- How will you validate your YAML's for best practices



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Course Outline

- Managing Applications using Deployments
 - Understanding Deployments & ReplicaSet in Kubernetes
 - Understanding self-healing, scaling
 - **LAB:** Deploy a web application using deployment
 - Understanding need of services in Kubernetes
 - Type of services - ClusterIP, NodePort, LoadBalancer
 - Understanding relation between label of pod and selectors of service
 - **LAB:** Deploy a web application using deployment and expose it with service
 - Understanding Manual Scaling of Pods
 - **LAB:** Scaling Deployment manually
 - Understanding Horizontal Pod Autoscaler (HPA)
 - **LAB:** Creating HPA & attach it with deployment with scaling policies configured
 - Understanding different deployment strategies
 - **LAB:** Upgrading Application using rolling update strategy
 - Checking health of application using Probes
 - Startup vs Readiness vs Liveness probes
 - **LAB:** Configure a deployment to use HPA and probes for health check of application
 - Understanding Daemonsets
 - **LAB:** Deploy log-collector fluentd in your cluster as daemonset

Troubleshooting, discussions & production issues:



- Application load has increased but HPA is not creating new pods
- Pods restart automatically and all configuration seems good
- Service is running but clients are not getting response from application
- Clients complain that service is pointing to different application
- Clients complaining that healthy pods are marked as unhealthy
- During heavy workload, healthy containers are marked as unhealthy
- Pod status is "running" but application is not working
- Why nodePort service is not an ideal solution in production

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Course Outline

- Influence Pod Scheduling
 - How scheduler work
 - Running pod on specific nodes
 - Understanding nodeSelector
 - **LAB:** Using nodeSelectors in pod scheduling
 - Affinity and Anti-affinity in pod scheduling
 - **LAB:** Deploying pods with required node affinity
 - **LAB:** Deploying pods with preferred node affinity
 - Advanced Affinity options in pod scheduling
 - **LAB:** Multiple affinity selection
 - **LAB:** Using not equal to operators in affinity selection
 - Understanding taints & tolerations
 - **LAB:** Applying taints on nodes
 - **LAB:** Scheduling pods with tolerations
 - Understanding pod priority classes
 - **LAB:** Creating pod priority classes
 - **LAB:** Scheduling pods with priority class
 - Marking nodes under maintenance
 - **LAB:** Configure node not to run any more workload

Troubleshooting, discussions & production issues:



- Pod remain in pending state, even when resources are available in cluster
- Your business application pods are not deploying on a specific node
- Pods are giving error of failed-scheduling
- You have a high priority application currently in pending state. You want scheduler to give preference to this application for allocating resources

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Course Outline

- Understanding Secrets and ConfigMap
 - Understanding need of secrets and types of secrets
 - **LAB:** Creating secrets using imperative and declarative way
 - **LAB:** Using secrets in pods to supply sensitive data to application
 - Changing behaviour of applications using ConfigMap
 - **LAB:** Creating ConfigMap
 - **LAB:** Using ConfigMap in Deployment

Troubleshooting, discussions & production issues:



- Pod CrashLoopBackOff just after starting
- Pod giving CreateContainerConfigError after deployed on node
- You have created configMap but pod is showing error configMap not-found
- You are getting a mis-configured configMap error

- Namespaces in Kubernetes
 - Understanding need of Namespaces
 - **LAB:** Creating Namespaces
 - Understanding Resource Quotas
 - **LAB:** Attaching resource quota to a namespace
 - Understanding LimitRange
 - **LAB:** Attaching LimitRange to a namespace
 - Controlling communication using Network Policies
 - **LAB:** Limiting pod communication using network policies

Troubleshooting, discussions & production issues:



- Pod getting un-scheduleable error when deploying in a namespace
- Discussing admission controllers - validating vs mutating
- You are getting unauthorized error when trying to deploy pod in a namespace
- When deleting a namespace its stuck in Terminating state



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Course Outline

- Storage in Kubernetes & Helm Charts
 - Understanding the Need of Persistent Storage in Kubernetes
 - Understanding persistent volumes, persistent volume claims and storage classes
 - **LAB:** Deploy a NFS server for shared storage
 - **LAB:** Deploying application to use static storage to store application data
 - Understanding Helm Charts
 - **LAB:** Integrating Helm with Kubernetes cluster
 - **LAB:** Deploying web application using Helm with custom values
 - **LAB:** Deploying prometheus and grafana using Helm
 - Understanding dynamic storage using storage classes
 - **LAB:** Installing nfs provisioner and storage class using Helm
 - **LAB:** Deploying application configured to use storage class

Troubleshooting, discussions & production issues:



- PVC is stuck in pending mode even after connecting with PV
- PV is not able to connect to a new PVC and PV status is showing released
- Deployment in dev namespace not able to connect to PVC
- Your PVC request is not accepted by the PV showing binding error
- How can you create a template from a Helm chart and use it

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Course Outline

- Role Based Access Control in Kubernetes
 - Understanding the need of RBAC in Kubernetes
 - Understanding Kubernetes roles and cluster-roles
 - **LAB:** Creating Users in Kubernetes
 - **LAB:** Configuring access to users using roles and cluster-roles
 - Understanding service accounts in Kubernetes
 - **LAB:** Using a custom service account to run pods

Troubleshooting, discussions & production issues:



- User trying to create pod in dev namespace is getting forbidden error
- How can a user check what he/she is allowed to do in a namespace
- Pod is getting in-correct service account error when trying to deploy
- How you can disable auto-mount service-account token for service account

- Backup Restore and Upgrading Cluster
 - Kubernetes components to backup
 - Understanding native backup tools and third party applications
 - **LAB:** Backup cluster using etcdctl / etcdutl
 - **LAB:** Restore cluster using etcdctl / etcdutl
 - Understanding cluster upgrade
 - **LAB:** Upgrade Kubernetes cluster using kubeadm

Troubleshooting, discussions & production issues:



- How will you minimise api-version mismatch issues while upgrading cluster

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Course Outline

- Ingress and Gateway API
 - Understanding need to Ingress controllers and Ingress
 - Understanding Gateway API
 - **LAB:** Installing ingress controller using Helm
 - **LAB:** Configuring ingress resource to manage traffic
 - **LAB:** Using a Gateway API to control incoming traffic

Troubleshooting, discussions & production issues:



- You are getting "No endpoints available" or 503 errors when connecting
- Ingress controller is working good. Ingress is properly configured still you are getting 502 Bad Gateway errors
- User browser's are giving security warnings or connection refused errors with TLS

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Troubleshooting, discussions & production issues:



- How will you minimise api-version mismatch issues while upgrading cluster

- **Preparing for the Certified Kubernetes Administrator Exam**
 - Mock Exam for Certified Kubernetes Administrator